Chapter test

5 Metabolism and enzymes

Name: _____ ( ) Class: _____ Date: _____

Total marks: 40 marks
Time allowed: 40 minutes
Answer ALL questions.

Score: / 40

A. Concept checking (10 marks, 1 mark each)

1. Which of the following is/are catabolic reaction(s)?

   (1) amino acids → polypeptide + water
   (2) triglyceride → fatty acids + glycerol
   (3) glucose + oxygen → carbon dioxide + water + energy

   A. (1) only
   B. (2) only
   C. (1) and (2) only
   D. (2) and (3) only ☐

2. Which of the following is ***not*** a property of enzymes?

   A. Enzymes are not consumed in the reaction.
   B. A small amount of enzymes is enough to catalyse a reaction.
   C. Enzymes increase the activation energy of the reaction.
   D. Enzymes can only work on one or a few substrates. ☐

★ 3. 1 cm^3 of 0.3% amylase solution is added to 1 cm^3 of 1% starch solution. After 10 minutes, the mixture is tested with iodine test, Benedict's test and a protein test strip. What would be the possible result?

	Iodine test	*Benedict's test*	*Protein test strip*
A.	positive	negative	positive
B.	negative	positive	positive
C.	positive	positive	negative
D.	negative	negative	negative

☐

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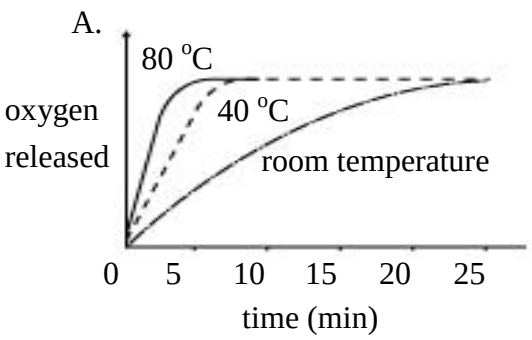
4. Which of the following properties of enzymes can be explained with the 'lock-and-key' model?

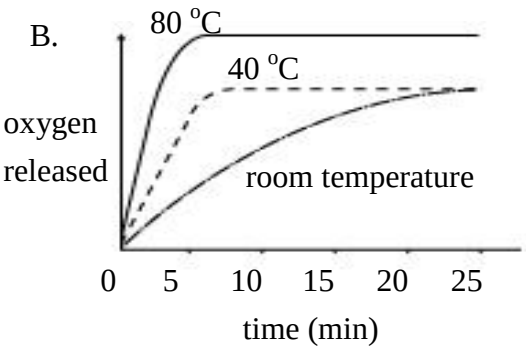
   A. Enzymes speed up chemical reactions.
   B. Enzymes are protein molecules.
   C. Enzymes are specific in action.
   D. Enzymes work best at their optimum conditions. ☐

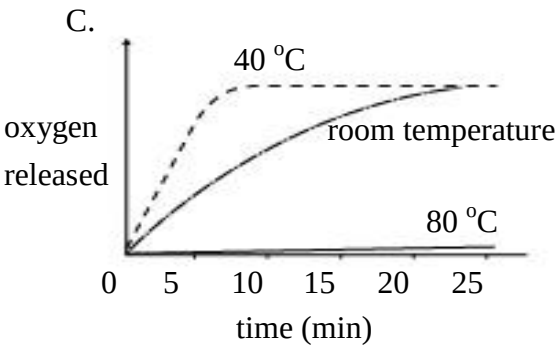
★ 5. In an experiment, a student accidentally added additional 10 mL of substrate solution into the reactants containing enzymes. He found that the rate of reaction has changed. What is the possible change and explanation for the change?

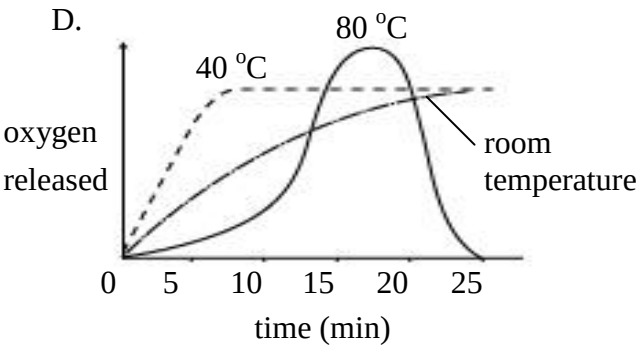
   A. The rate of reaction decreased because the reactants were diluted by the extra substrate solution.
   B. The rate of reaction increased because the substrates collide more frequently with the enzymes.
   C. The rate of reaction decreased because the enzyme-to-substrate ratio decreased.
   D. The rate of reaction increased because there are more active sites to bind with more substrates. ☐

★ 6. A test tube containing 5 cm^3 of hydrogen peroxide and 2 cm^3 of 1% catalase solution was put at room temperature. The amount of oxygen released was recorded at 5-minute intervals. The experiment was repeated at 40 °C and 80 °C. Which of the following graphs correctly shows the possible result of the three conditions?

A.


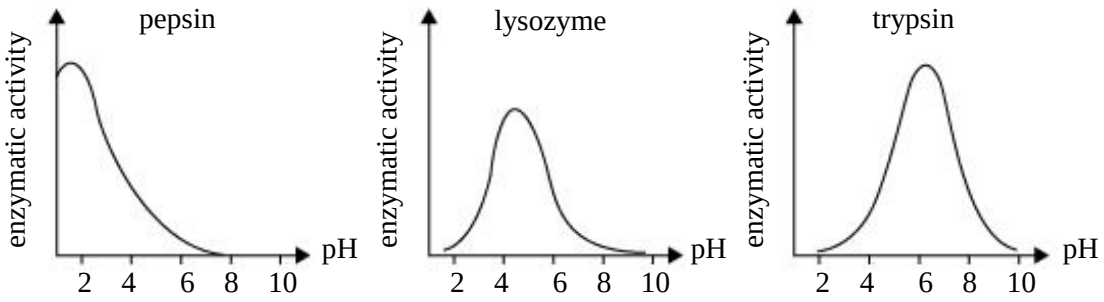
B.


C.


D.


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★ 7. The graphs below show the enzymatic activity of three enzymes which can be found inside the human body.



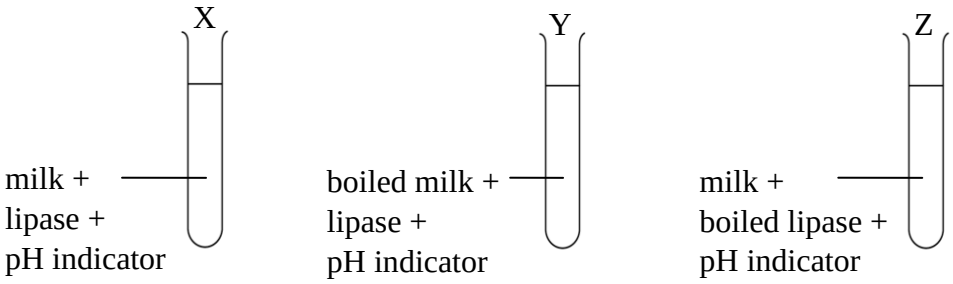
Which of the following is/are correct?

(1) At pH 8, all trypsin would be denatured.
(2) At pH 4, lysozyme has the highest enzymatic activity among the three enzymes.
(3) Pepsin is active in the human stomach.

A. (1) only
B. (2) only
C. (1) and (2) only
D. (2) and (3) only

☐

★ 8. Lipase can catalyse the breakdown of fat. In an experiment, lipase and pH indicator are added to full cream milk containing 3.5% fat. Consider the three test tubes as shown below.



The pH value of the contents of each tube was measured at the beginning and after 30 minutes. What changes in pH value would be observed?

	Tube X	***Tube Y***	***Tube Z***
A.	decreased	decreased	decreased
B.	increased	increased	increased
C.	increased	increased	no change
D.	decreased	decreased	no change

☐

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9. Which of the following descriptions about enzyme inhibitors is ***incorrect***?

A. Inhibitors cause a decrease in the rate of enzymatic reactions.
B. All inhibitors block the entry of substrates into the active sites.
C. Cyanide and mercury are examples of enzyme inhibitors.
D. Some inhibitors are used by humans as drugs to treat diseases. ☐

★ 10. The table below shows four choices of enzymes to be included in biological washing powder, and their optimal conditions.

Enzyme	Optimal temperature (°C)	Optimal pH
P	20 – 32	7.0 – 9.0
Q	23 – 24	5.0
R	90 – 95	2.0 – 2.5
S	39 – 41	10.0

Which enzyme is the most suitable to be used in the washing powder?

A. enzyme P
B. enzyme Q
C. enzyme R
D. enzyme S ☐

Question	1	2	3	4	5	6	7	8	9	10
Related section	5.1	5.2	5.3	5.3	5.4	5.4	5.4	5.4	5.4	5.5

B. Problem solving (12 marks)

★ 11. Catalase is an enzyme which catalyses the breakdown of hydrogen peroxide, producing oxygen and water. Peter conducted an investigation to study the activity of catalase in potato tissues as follows:

(1) Three potato cylinders (A, B and C) of 5 cm long are obtained from a potato tuber.
(2) Potato cylinder A is totally immersed in a boiling tube containing 10 cm^3 of hydrogen peroxide solution.
(3) The amount of gas released is collected for 5 minutes.
(4) Potato cylinder B is cut into 10 equal discs. Potato cylinder C is boiled for 5 minutes.
(5) Steps 1 to 3 are repeated by replacing potato cylinder A with B and C respectively.

The table on the next page shows the results.

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Treatment		Volume of gas collected (cm^3)
A	fresh potato cylinder + 10 cm^3 of hydrogen peroxide solution	15
B	10 potato discs + 10 cm^3 of hydrogen peroxide solution	25
C	boiled potato cylinder + 10 cm^3 of hydrogen peroxide solution	0

(a) Explain the results of treatments A and B. (3 marks)

(b) Explain the results of treatments A and C. (3 marks)

(Total: 6 marks)

★ 12. Jelly contains a type of protein called gelatin, which makes jelly solution solidify when cooled. Kate prepared two bowls of jelly solution (P and Q).

- P was prepared by dissolving a pack of jelly powder in 40 ^oC water.
- Q was prepared by dissolving a pack of jelly powder in 40 ^oC water and then mixed with fresh pineapple pieces.

Kate put both bowls in the refrigerator for three hours. She found that only one bowl solidified.

(a) Suggest which bowl of jelly solution failed to solidify. Explain your answer. (2 marks)

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(b) Kate wants to make pineapple jelly. She proposes the following two ways to make pineapple jelly. Explain whether each of her suggestion would work or not.

(i) Extract juice from the pineapple pieces and then mix the juice with the jelly solution (2 marks)

(ii) Boil the pineapple pieces first before adding them to the jelly solution (2 marks)

(Total: 6 marks)

C. Exam challenge (18 marks)

★★ 13. Sucrase catalyses the digestion of sucrose to produce glucose and fructose.



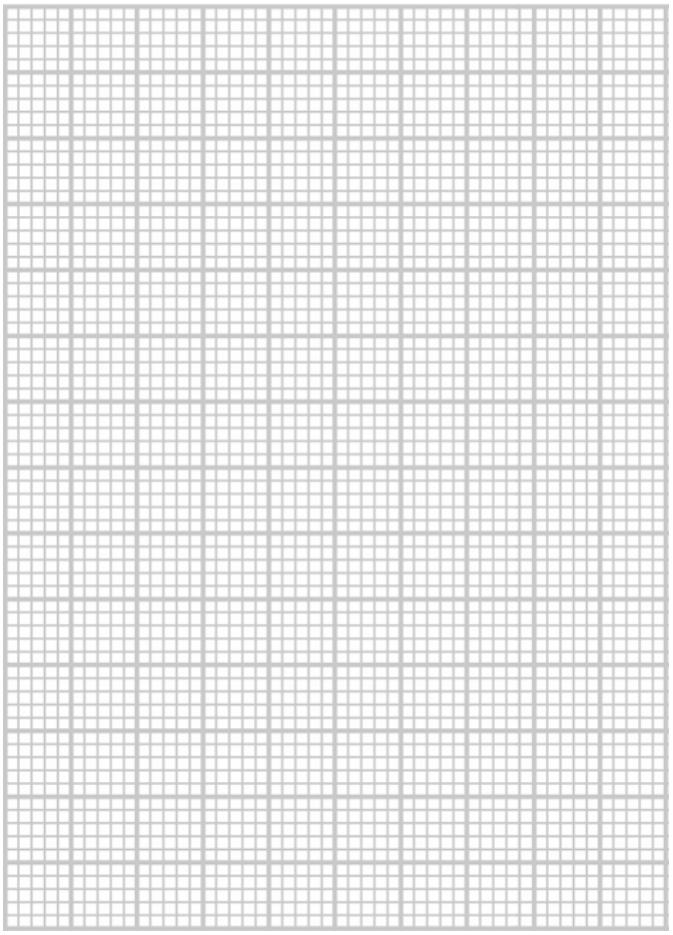
$$\text{sucrose} \xrightarrow{\text{sucrase}} \text{glucose} + \text{fructose}$$

A student carried out an experiment to study the effect of increasing the concentration of sucrase on the rate of this reaction. He used seven different concentrations of sucrase, while keeping the concentration of sucrose constant. For each sample, the time required for complete digestion of sucrose was measured. All the reactions were carried out at 40 ^{o}C. The results are shown in the table below.

Sucrase concentration (%)	Time required to digest the sucrose (seconds)
0.2	560
0.4	420
0.6	320
0.8	220
1.0	150
1.2	120
1.4	100

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(a) (i) Plot a graph to show the effect of sucrase concentration on the rate of sucrose digestion. (4 marks)



(ii) With reference to your graph, describe and explain the effect of increasing the concentration of sucrase on the rate of sucrose digestion. (3 marks)

(b) Explain why all the reactions were carried out at 40 ^{o}C. (1 mark)

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(d) Suggest how the temperature can be kept constant at 40 ^{o}C. (1 mark)

(Total: 9 marks)

★★ 14. After an apple is cut open and exposed to oxygen in the air, the cut surface of the apple will turn brown.

(a) Explain the observations in the following two cases.

(i) If a cut apple is dipped into vinegar, the cut surface does not turn brown. (2 marks)

(ii) If an apple is refrigerated immediately after cut, the cut surface turns brown slightly. (2 marks)

(b) Design an experiment to investigate the effect of pH on the colour change on the cut surfaces of apple pieces. (5 marks)

(Total: 9 marks)

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Marking scheme

A. Concept checking (10 marks)

1.	D	2.	C	3.	B	4.	C	5.	B
6.	C	7.	D	8.	D	9.	B	10.	A

(1 mark each)

B. Problem solving (12 marks)

11. (a) The total surface area of the ten potato discs was greater than that of a potato cylinder. (1)
In B, the surface area for catalase to act on hydrogen peroxide was greater than in A, (1)
so the rate of reaction of B was higher than that of A. (1)

(b) Boiling caused the catalase in the potato cylinder to become denatured. (1)
The denatured catalase could not catalyse the breakdown of hydrogen peroxide, (1)
so no gas was released in C. (1)

(Total: 6 marks)

12. (a) Q failed to solidify. (1)
Pineapple contains proteases which can catalyse the breakdown of gelatin. (1)

(b) (i) The jelly solution would fail to solidify. (1)
The pineapple juice contains proteases which are still functional outside of cells. (1)

(ii) The jelly solution would solidify. (1)
Boiling would cause the proteases in pineapple to become denatured. (1)

(Total: 6 marks)

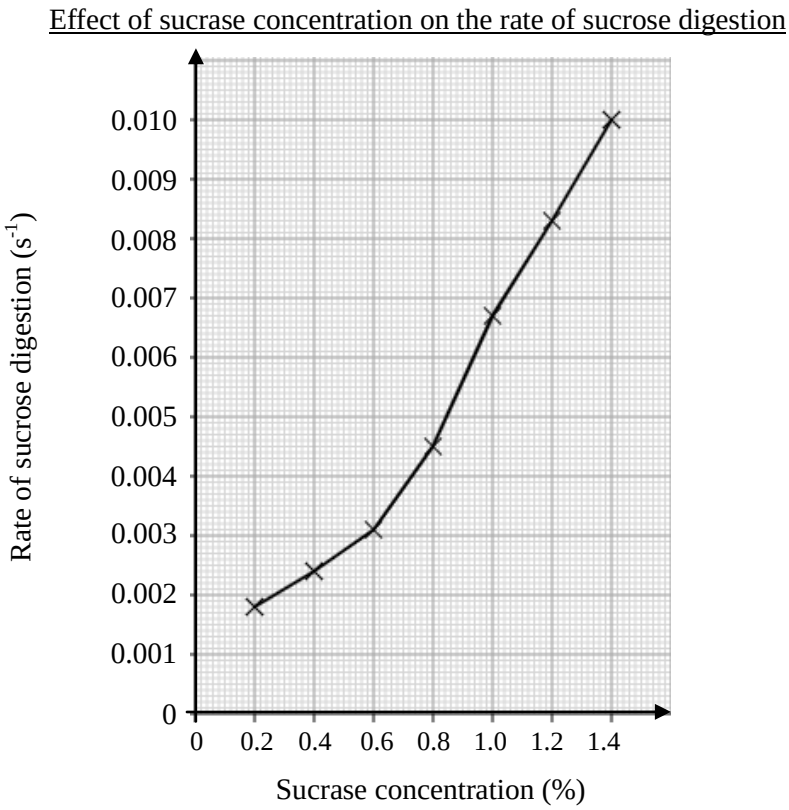
C. Exam challenge (18 marks)

13. (a) (i) Calculate the rate of sucrose digestion using the time required to digest the sucrose

Time required to digest the sucrose (seconds)	**Rate of sucrose digestion (s^{-1})**
560	0.0018
420	0.0024
320	0.0031
220	0.0045
150	0.0067
120	0.0083
100	0.0100

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Correct title (1); correct choice of axes (1); axes with labels and units (1); correct plotting of points (1)



(ii) The rate of sucrose digestion increases as the concentration of sucrase increases. (1)
More active sites are available to bind with sucrase. (1)
More enzyme-substrate complex can be formed per unit time. (1)

(b) To minimise the effect of temperature on enzyme action. (1)

(c) The reaction mixtures are kept in a water bath of 40 ^{o}C throughout the experiment. (1)

(Total: 9 marks)

14. (a) (i) Vinegar has low pH / is acidic. (1)
Enzymes involved in the browning process are denatured by low pH. (1)

(ii) Low temperature decreases the enzyme activity. (1)
Enzymes involved in the browning process are inactivated and the browning process occurs at a slower rate. (1)

(b) Cut an apple into cubes of the same size. (1)
Immerse the apple cubes into solutions of varying pH (minimum 3 different pH). (1)
The apple cubes are treated under same conditions, e.g. temperature, volume of pH solution, etc. (1)
Compare colour changes after a fixed period of time. (1)
Repeat the experiment to obtain another set of data. (1)

(Total: 9 marks)

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